

# NetExam

Sri Lanka Institute of Information Technology

What is the verb, which we use to indicate Create operation? (1) GET (2) POST (3) PUT (4) DELETE

Select one:

- ☒ a. Only (2)
- ☐ b. Both of them
- ☐ c. None of them
- ☐ d. Only (3)

What are the advantages of MVC?

Select one:

- ☒ a. Presentation is separated from the business logic
- ☐ b. All of the others
- ☐ c. Event-handling is separated from business logic
- ☐ d. Presentation is separated from Event-handling

13

answered

out of

question

The basic architecture of the distributed systems is called

Select one:

- ☒ a. Client-server architecture
- ☐ b. N-tier architecture
- ☐ c. 3-tier architecture
- ☐ d. Service oriented architecture

Bootstrap is a?

Select one:

- ☐ a. Scripting language
- ☐ b. Plugin
- ☒ c. Framework
- ☐ d. Library

Which one of the following is **not based** on GIT?

Select one:

- ☐ a. SourceForge
- ☐ b. GitHub
- ☒ c. OneDrive
- ☐ d. GitLab

What are the benefits of web services?

1. Static integration
2. Standardized Protocol ✓
3. Interoperability ✓
4. Constant by design
5. Low Cost of communication ✓

HTTP, XML

Select one:

- ☐ a. 1 and 5
- ☐ b. 3, 4, and 5
- ☒ c. 2, 3, and 5
- ☐ d. 1, 2, and 4

Procedural programming focuses on

Select one:

- ☐ a. Flow control with GOTO statements
- ☐ b. Use of namespaces/packages
- ☐ c. Functions (as in functional programming)
- ☒ d. Modularity and code reuse

Procedural Programming කියන්නේ

☞ program එක procedures / functions වලට බෙදලා ලියන programming style එකක්

◊ Main Focus

☞ Code එක:

modules (functions) වලට බෙදනවා

reusability (නැවත භාවිතා කිරීම) වැඩි කරනවා

✓ main focus එක:

☞ Modularity + Code reuse

Which one of the following is not a purpose of having a VCS?

Select one:

- ☐ a. Restoring previous versions
- ☐ b. Backup
- ☒ c. Code formatting
- ☐ d. Collaboration



"In software runtime architecture we may need additional virtualization layers, depending on the type of the system." Between which layers we may need these virtualization layers?

Select one:

- ☒ a. Between Operating System and Hardware
- ☐ b. Between User and Application
- ☐ c. Between User and Hardware
- ☐ d. Between Application and Operating System

What is false about Micro-services Architecture?

Select one:

- ☐ a. The system is developed with many components
- ☒ b. Can use single database or multiple databases
- ☐ c. May use SOAP or REST for communication
- ☐ d. Made of many web services

## The difference between the Servlets and the JSP:

Select one:

- ☒ a. Servlets are used to develop web applications and JSP is used to develop web sites
- ☐ b. JSP is an application and Servlet is a web page
- ☐ c. Servlet can generate HTML and JSP can generate JAVA code
- ☐ d. JSP is a web page and Servlet is an application

Servlets සහ JSP (JavaServer Pages) දෙකම Java web technologies වුණත්  
☞ use ධූ

◊ Servlet

☞ Servlet කියන්නේ:

Java class එකක්

server-side logic (application logic) handle කරනවා

Form processing, request handling වගේ වැඩ කරනවා

◊ JSP (JavaServer Pages)

☞ JSP කියන්නේ:

web page එකක් වගේ

HTML + Java mix කරලා ලියනවා

✓ UI (presentation) create කරන්න use කරනවා

◊ Options explain කරමු

→ දෙකම web apps වල use වෙනවා

→ වැරදි definition

→ ප්‍රති (reverse) statement

→ සරි explanation එක

The key technique in RESTful communication uses: (1) URL (2) HTTP Verbs

Select one:

- ☒ a. Only (1)
- ☐ b. None of them
- ☐ c. Only (2)
- ☒ d. Both of them

RESTful communication වල main technique දෙකක් තියෙනවා:

☒ 1. URL (Uniform Resource Locator)

☞ Resources identify කරන්න use කරනවා

✓Example:

/users

/users/1

☒ 2. HTTP Verbs (Methods)

☞ Action එක කියන්න use කරනවා

✓Examples:

GET → read

POST → create

PUT → update

DELETE → delete

Why do centralized control in distributed systems useful to engineers?

Select one:

- ☐ a. Maintaining the system is easier
- ☒ b. All of the others
- ☐ c. Easier to update
- ☐ d. Easier to modify and extend

The communication techniques in distributed systems can be categorized as

Select one:

- ☐ a. RPC, RMI, and CORBA
- ☒ b. Functional oriented, Message oriented, and Resource oriented
- ☐ c. RPC/RMI, SOAP, and DC
- ☐ d. Delta-Communication, SOAP communication, and RESTful communication



Which one is the true statement regarding systems with web-services?

Select one:

- ☒ a. Supports dynamic integration of components
- ☐ b. Use proprietary protocols for communication
- ☐ c. Web services always use monolithic architecture
- ☐ d. Web services are about user to program interaction

Web services කියන්නේ

☞ systems අතර dynamic integration (flexible connection) කරන්න use කරන technology එකක්

☒ a. Supports dynamic integration of components

✓සහි

☞ Different systems / components

☞ dynamically connect වෙලා work කරන්න පුළුවන්

Programming languages provide

Select one:

- ☐ a. Semantics and functions
- ☐ b. Syntax and procedures
- ☐ c. Syntax and functions
- ☒ d. Syntax and semantics

Programming languages provide කරන basic components දෙක:

☒ 1. Syntax (වි)

☞ Code ලියන rules / structure  
✓

if (x > 0) { } වගේ format එක

☒ 2. Semantics (අර්ථය)

☞ Code එකේ meaning (කියාදෙන දේ)  
✓

$x = x + 1 \rightarrow x$  එක increase කරනවා



Identify correct ways to maintain the quality of the code:

1. Documenting the code
2. Functional testing
3. Regular code reviews
4. Clear requirements

Select one:

- ☐ a. Only (2) and (3)
- ☐ b. Only (1) and (3)
- ☐ c. Only (2) and (4)
- ☒ d. All (1), (2), (3), and (4)

Refactoring is one of the aspect of quality code. What does it mean?

Select one:

- ☒ a. Re-arrange the code to improve efficiency
- ☐ b. Manage the configurations
- ☐ c. Extending the code
- ☐ d. Test and fix the bugs

What is the operation for HTTP verb PUT? (1) Create (2) Update

Select one:

- ☐ a. None of them
- ☐ b. Only (1)
- ☐ c. Only (2)
- ☒ d. Both of them

Explanation (සිංහලෙන්)

☞ HTTP PUT method එක use වෙන්නේ:

☒ Update (use එක)

☞ existing resource එක update කරන්න

☒ Create (සමහර අවස්ථාවල)

☞ resource එක already තැන්පත්

☞ create කරන්නත් පුළුවන්

✓ PUT = create or replace)

The resource is determined by? (1) The domain name (2) CRUD operations (3) URL segment

Select one:

- ☐ a. None of them
- ☒ b. Only (3)
- ☐ c. Only (2)
- ☐ d. Only (1)

Some applications developed using particular programming languages can be directly run on the OS with

Select one:

- ☒ a. This statement is true
- ☐ b. Cannot validate the statement without knowing the OS
- ☐ c. This statement is not conclusive
- ☐ d. This statement is false

සමහර programming languages වලින් develop කරන applications  
↳ directly Operating System (OS) මත run ✓

◇ Example

C / C++ programs

↳ compile කරලා

↳ direct machine code එකක් ලෙස OS එකේ run වෙනවා

What are the example for data representational elements in HTML?

Select one:

- ☐ a. Tables, Input
- ☒ b. Lists, Tables
- ☐ c. Input, Buttons
- ☐ d. Buttons, Tables

Explanation (සිංහලෙන්)

☞ Data representational elements කියන්නේ

☞ data display / represent කරන්න use කරන HTML elements

☒ Lists

☞ Data list එකක් ලෙස show කරන්න

✓ul>, <ol>, <li>

☒ Tables

☞ Structured data (rows & columns) show කරන්න

✓table>, <tr>, <td>

What is used by Maven to store project details

Select one:

- ☐ a. CDN
- ☐ b. VCS
- ☒ c. XML file
- ☐ d. JSON file

Maven project එකේ details store කරනවා:

☞ pom.xml file එක තුළ

◊ What is inside pom.xml?

☞ Project information:

dependencies

plugins

build settings

project name, version

◊ Why XML?

☞ Maven uses XML format

☞ structured way එකකින් data define ✓



Which of the following is incorrect about JAX-RS annotations?

Select one:

- ☐ a. PATH - Relative path of the resource class/method
- ☐ b. POST - Create a new resource
- ☒ c. PUT - Used to add resources
- ☐ d. HEAD - Get status about method availability



The CRUD operations are determined by?

1. Read & Create

2. PUT & DELETE

3. HTTP verb ~~X~~

4. All of them

CRUD operations (Create, Read, Update, Delete)

↳ REST APIs වල determine කරන්නේ:

↳ HTTP verbs (methods)

◇ Mapping එක

Create → POST

Read → GET

Update → PUT

Delete → DELETE

↳ මේවා HTTP verbs

Select one:

- ☒ a. Only (3)
- ☐ b. Only (4)
- ☐ c. Only (2)
- ☐ d. Only (1) and (2)

What is not related to the scalability?

Select one:

- ☐ a. Distribution of the components
- ☒ b. Testing
- ☐ c. Number of users
- ☐ d. Concurrent access

What modules are implemented on the server-side of BAW-MVC?

1. Component
2. Controller
3. View
4. Model

Select one:

- ☒ a. Only (1) and (2)
- ☐ b. Only (1), (2), and (3)
- ☐ c. Only (3), and (4)
- ☐ d. All of them

BAW-MVC (Browser–Application–Web MVC) architecture එකේ  
☞ server-side (Application layer) එකේ නියමිතේ:

☒ 1. Component

☞ Business logic handle කරන ✓

☒ 2. Controller

☞ Requests handle කරලා

☞ correct response return කරනවා

Which one of the following is not a benefit of CDN?

Select one:

- ☐ a. Reducing bandwidth costs
- ☐ b. Improving the security of the system
- ☒ c. Support collaboration
- ☐ d. Improving system's loading time

CDN (Content Delivery Network) කියන්නේ

☞ content (images, videos, files) fast deliver කරන්න use කරන system එකක්

◇ Benefits of CDN

☒ a. Reducing bandwidth costs

☞ Data caching නිසා bandwidth usage අඩුවෙයි

☒ b. Improving the security of the system

☞ DDoS protection, HTTPS support වගේ features ✓

Which one of the following is not a VCS model?

Select one:

- ☐ a. Centralized version control systems
- ☐ b. Local version control systems
- ☐ c. Distributed Version Control Systems
- ☒ d. Global version control systems

Which language supports both procedural and object oriented programming?

Select one:

- ☐ a. Java
- ☐ b. C
- ☒ c. C++
- ☐ d. C#

Which is NOT correct about the View development in web-based applications?

Select one:

- ☐ a. View is deployed to the client's computer
- ☐ b. View should be loaded to the client's browser
- ☐ c. Server-side code in View may execute in server-side
- ☒ d. View should communicate with model to load the data from database

MVC architecture එකේ responsibilities clearly separate වෙලා තියෙනවා

◇ Correct flow

→ Controller → Model → Database

☞ View directly Model එකේක communicate ❌

◇ Options explain කරමු

a. View is deployed to the client's computer ☑

☞ browser එකේ run වෙන නිසා correct

b. View should be loaded to the client's browser ☑

☞ UI browser එකට load වෙනවා

c. Server-side code in View may execute in server-side ☑

☞ JSP වගේ technologies වල possible

d. View should communicate with model ❌

☞ MVC rules break වෙනවා



Which cannot be used to develop the server-model?

Select one:

- ☐ a. PHP
- ☒ b. HTML
- ☐ c. Node.js
- ☐ d. JAVA

What are the languages, which can be used to write code in an HTML page, which can

Select one:

- ☐ a. CSS, Java
- ☐ b. None of the above
- ☐ c. JS, Java
- ☒ d. CSS, JS

What is **NOT** a correct statement about CSS?

Select one:

- ☐ a. None of the others
- ☐ b. Eliminate HTML formatting tags
- ☐ c. Save work and time by reusing style sheets
- ☒ d. Slow down loading

What HTTP verb is used for Read operation?

- (1) GET
- (2) POST
- (3) PUT
- (4) DELETE

Select one:

- ☐ a. (1) and (2) only
- ☐ b. All of them
- ☒ c. Only (1)
- ☐ d. only (2)

## Evaluate WSDL

Select one:

- ☒ a. All of others
- ☐ b. WSDL helps in dynamic service integration
- ☐ c. WSDL is a contract between the Web service and the client
- ☐ d. WSDL describes a web service

WSDL (Web Services Description Language) කියන්නේ  
☞ web service එකක් describe කරන XML-based language එකක්

◊ Options explain කරමු

☒ b. WSDL helps in dynamic service integration

✓ කරන්න පුළුවන්

☒ c. WSDL is a contract between the Web service and the client

☞ එකේ methods, inputs, outputs define කරනවා

☒ d. WSDL describes a web service

✓ එකේ full description එක දෙන්නෙ WSDL

Which of the following is not correctly defining loosely coupled architecture of web service?

Select one:

- ☐ a. services can be independently updated
- ☐ b. software systems become more manageable
- ☐ c. allows integration between different systems
- ☒ d. Low scalability

In the Model-View-Controller (MVC) architectural style, the model defines the

select one:

- ☐ a. Interfaces
- ☐ b. Presentation
- ☒ c. Business-logic
- ☐ d. Communication model

→ presentation (UI)

Controller ✗ කරනවා

\*There is a gap between development and implementation environments, such as having different platforms.

Which one of the following helps us to overcome these issues?

Select one:

- ☐ a. Version controlling
- ☐ b. CDN
- ☐ c. Dependency management
- ☒ d. Virtualization

Development environment එක සහ deployment (implementation) environment එක අතර  
↳ difference (OS, platform, configs) තියෙනවා නම්  
↳ gap එක solve කරන්න use කරන best solution එක:

👍 ✓

◇ Why Virtualization?

↳ Same environment එක simulate කරන්න පුළුවන්  
↳ (e.g., Virtual Machines, Containers)

"It works on my machine" problem එක fix කරනවා  
Consistent environment එකක් ලැබෙනවා



Which of the following is not a software category?

Select one:

- ☐ a. CASE tools
- ☐ b. Utilities
- ☐ c. System software
- ☒ d. Cloud computing

CSS can be written as

Select one:

- ☒ a. all of other answers
- ☐ b. Inline CSS
- ☐ c. External CSS sheets
- ☐ d. Internal CSS sheets